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SUPERSEDING
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FEDERAL SPECIFICATION

TUBE, NONMETALLIC; POLYVINYL CHLORIDE (PVC), FLEXIBLE (FOR LABORATORY AND MEDICAL USE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers clear polyvinyl chloride (PVC) tubing for laboratory and medical use.

1.2 Classification.

1.2.1 Size and wall thickness. Plastic tubing shall be the following nominal sizes and wall thickness (see table I and 6.2).

TABLE I. Nominal size, diameter, and thickness of tubes, inch

Nominal size and inside diameter 1/	Outside diameter	Wall thickness 2/
1/16	1/8	1/32
1/16	3/16	1/16
3/32	5/32	1/32
3/32	7/32	1/16
1/8	3/16	1/32
1/8	1/4	1/16
5/32	7/32	1/32
5/32	9/32	1/16
3/16	1/4	1/32
3/16	5/16	1/16
3/16	3/8	3/32
3/16	7/16	1/8
3/16	9/16	3/16
1/4	5/16	1/32
1/4	3/8	1/16
1/4	7/16	3/32
1/4	1/2	1/8
5/16	7/16	1/16
5/16	1/2	3/32
5/16	9/16	1/8
3/8	1/2	1/16
3/8	9/16	3/32
3/8	5/8	1/8
7/16	9/16	1/16
7/16	5/8	3/32
7/16	11/16	1/8
1/2	5/8	1/16
1/2	11/16	3/32
1/2	3/4	1/8
1/2	13/16	5/32
9/16	3/4	3/32
9/16	13/16	1/8
9/16	7/8	5/32
5/8	13/16	3/32

1/ Tolerance in inside diameter for all sizes shall be plus or minus 1/64 inch.
2/ Tolerance on wall thickness for all sizes shall be minus nothing or plus 1/64 inch.

1.2.2 Length and form. Length and form shall be in accordance with 3.2.2 (see 6.2).

FSC 4720

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-775 - Hose, Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles and Strainers, Packaging of.

Military Standards:

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D638 - Tensile Properties of Plastics.
D746 - Brittleness Temperature of Plastics and Elastomers by Impact.
D792 - Specific Gravity and Density of Plastics by Displacement.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

United States Pharmacopeial Convention, Inc.:

Pharmacopeia of the United States (USP).

(Application for copies should be addressed to the Mack Printing Company, 20th and Northampton Streets, Easton, PA. 18042.)

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REQUIREMENTS

3.1 General. Tubing in accordance with this specification shall be colorless, clear, flexible polyvinyl chloride (PVC) capable of withstanding steam sterilization for 30 minutes.

3.2 Dimensions.

3.2.1 Size (inside diameter) and thickness. Size (inside diameter) and thickness of tubes shall be in accordance with table I (see 6.2).

3.2.2 Length and form. Length of tubes shall be as specified (see 6.2). Tubes shall be in the form of coils, reels, or in straight lengths, as specified (see 6.2). When coils or reels are required, they shall be furnished in 50, 100, 250, 500, or 1000 foot lengths as specified.

3.3 Flexibility. Tubing shall be flexible but sufficiently rigid to prevent collapse during use (see 4.4.1).

3.4 Physical properties. Physical properties of tubes shall be in accordance with table II (see 4.4.2).

TABLE II. Physical properties

Property	Requirement	Test method (ASTM)
Brittleness temperature, °C	-46	D746
Specific gravity	1.15-1.80	D792
Tensile strength, p.s.i.	1800-2100	D638
Elongation, percent	300 min.	D638

3.5 Chemical resistance. Tubing shall not discolor, flake, crack, or pit in the presence of strong and weak nonoxidizing acids and weak alkalies (see 4.4.3).

3.6 Sterilization. Tubing shall be capable of sterilization. Surfaces of tubes shall not crack or become etched when tested in accordance with 4.4.4.

3.7 Heat resistance. Tubes shall not collapse when tested in accordance with 4.4.5.

3.8 Safety. Tubes shall be safe (nontoxic) when tested in accordance with 4.4.6.

3.9 Metal content. When tested in accordance with 4.4.7, ash residue of tubes shall contain no more than 5 parts per million (p.p.m.) of heavy metals such as lead.

3.10 Workmanship. Tubes shall be free from defects which mar their appearance or affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. A lot shall consist of all tubing of the same size offered for delivery at the same time.

4.2.2 Sampling for inspection. Select sample size for visual examination in accordance with level S-3 of MIL-STD-105, and level S-1 for tests.

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4.3 Inspection.

4.3.1 Visual examination. Examine tube for clarity, dimensional, and workmanship requirements. The acceptable quality level (AQL) shall be 6.5 percent defective.

4.3.2 Tests. Failure to pass any of the tests specified herein shall cause rejection of the lot.

4.4 Test methods.

4.4.1 Flexibility test. Wrap tubing around a mandrel of diameter equal to 5 times the O.D. of the tube being tested (see 3.3).

4.4.2 Physical properties test. Test specimens for physical properties in accordance with table III for conformance to 3.4.

TABLE III. Test methods for physical properties

Test for	ASTM Test Methods
Brittleness	D746
Specific gravity	D792
Tensile strength and elongation	D638

4.4.3 Chemical resistance test. Immerse test specimens for 24 hours in reagent solutions at a temperature of 40°C ± 1.7°C. The reagent solutions shall be sulfuric acid (30 percent), sulfuric acid (5 percent) and sodium hydroxide (10 percent). At end of 24 hour immersion, remove specimens and examine for conformance to 3.5.

4.4.4 Sterilization. Test specimens as follows (see 3.6).

(a) Prepare commercially cold sterilizing solutions, benzalkonium chloride USP.
(b) Boil solution.
(c) Autoclave at 15 p.s.i. for 30 minutes.

4.4.5 Heat resistance. Regulate temperature of a laboratory oven to 93 ± 2.7°C. Place sample specimens in the preheated oven for 1 hour (see 3.7).

4.4.6 Safety (toxicity) test. Tubes shall be tested for safety in accordance with the safety (toxicity) test method of USP for class II plastics (see 3.8).

4.4.7 Metal concentration. Accurately weigh a 5-gram sample into a 1-oz. Vycor crucible. Reduce to ash in a muffle furnace at 650°C for 8 hours. Let cool to room temperature and add 15 ml. of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil for 2 to 3 minutes, remove, let cool, and filter through Whatman #2 filter paper into a 50-ml. volumetric flask. Rinse the residue on the filter with deionized water and dilute to 50 ml. Prepare standard lead solutions of 1, 2, and 5 ppm. lead. Determine the absorbances of these solutions at the 283 millimicron wave length using a Hollow Cathode Pb lamp and a 10x scale expansion on a spectrophotometer, such as the Perkin-Elmer Model 303 Atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standards.

Calculation:

$$\frac{A \times F \times \text{dilution}}{W} = \text{ppm lead}$$

A = Absorbance
F = Factor = $\left(\frac{\text{Concentration standard in ppm Pb}}{\text{absorbance}}\right)$
W = Sample weight in grams

The calculated factors of the three standards are averaged to give ppm/unit A.

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4.5 Inspection of preparation for delivery. An inspection shall be made to determine whether the packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with Table IV. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

TABLE IV. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Any component missing, damaged or not conforming to the applicable specification or requirements stated herein.
Workmanship	Inadequate application components such as incomplete closure of container flaps, loose strapping or distortion of container.
Contents	Quantity per container is more or less than required. Gross weight exceeds requirements of the box specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. The tubing in quantities as specified (see 6.2), shall be packaged and packed in accordance with MIL-H-775. Packaging shall be level A or C and packing shall be level A, B, or C, as specified (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. Interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tubing is intended for use in medical and laboratory procedures.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size and wall thickness (see 1.2.1 and 3.2.1).
(c) Length and form required (see 1.2.2 and 3.2.2).
(d) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
(e) Quantity of tubing required (see 5.1.1 and 5.2.1).

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MILITARY CUSTODIAN:

Air Force - 82

Review interest:

DSA - CS

Coordinating activity:

DSA - CS

PREPARING ACTIVITY: GSA - FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
HEW - HSM - NIH
NASA - JFK
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